

Python Libraries for Data Science Simplified

The typical life cycle of a data science project involves jumping back and forth among various interdependent data science tasks using a range of tools, techniques, frameworks programming, etc. A data scientist typically needs to be involved in tasks like data wrangling, exploratory data analysis (EDA), model building and visualisation. Python provides almost all the necessary tools to effectively carry out these tasks, with dedicated libraries for each step.

Python comes with powerful libraries such as Pandas, Matplotlib, Plotly, Scikit-learn, etc, and advanced deep learning libraries such as TensorFlow, Keras, etc. Figure 1 explains a few widely used Python tools at different steps of a data science project.

Pandas, your buddy for Exploratory Data Analysis (EDA)

Key features

- Python version support: Officially Python 3.6.1 and above.
- Open source, fast, powerful, flexible and easy to use open source data analysis and manipulation.
- Built on top of the NumPy package, a lot of the structure of NumPy is used or replicated in Pandas. Data in Pandas is often used to feed statistical analysis in SciPy, plotting functions from Matplotlib, and machine learning algorithms in Scikit-learn.
- Before all of your fancy machine learning, you will have to

explore, clean and process your data thoroughly. For that, you need Pandas. Literally, you get a Microsoft Excel inside Python.

Tricks to try out!

The primary two components of Pandas are the Series and DataFrame. A Series is essentially a column and a DataFrame is a multi-dimensional table made up of a collection of Series.

```
# Creating a DataFrame
import pandas as pd
```

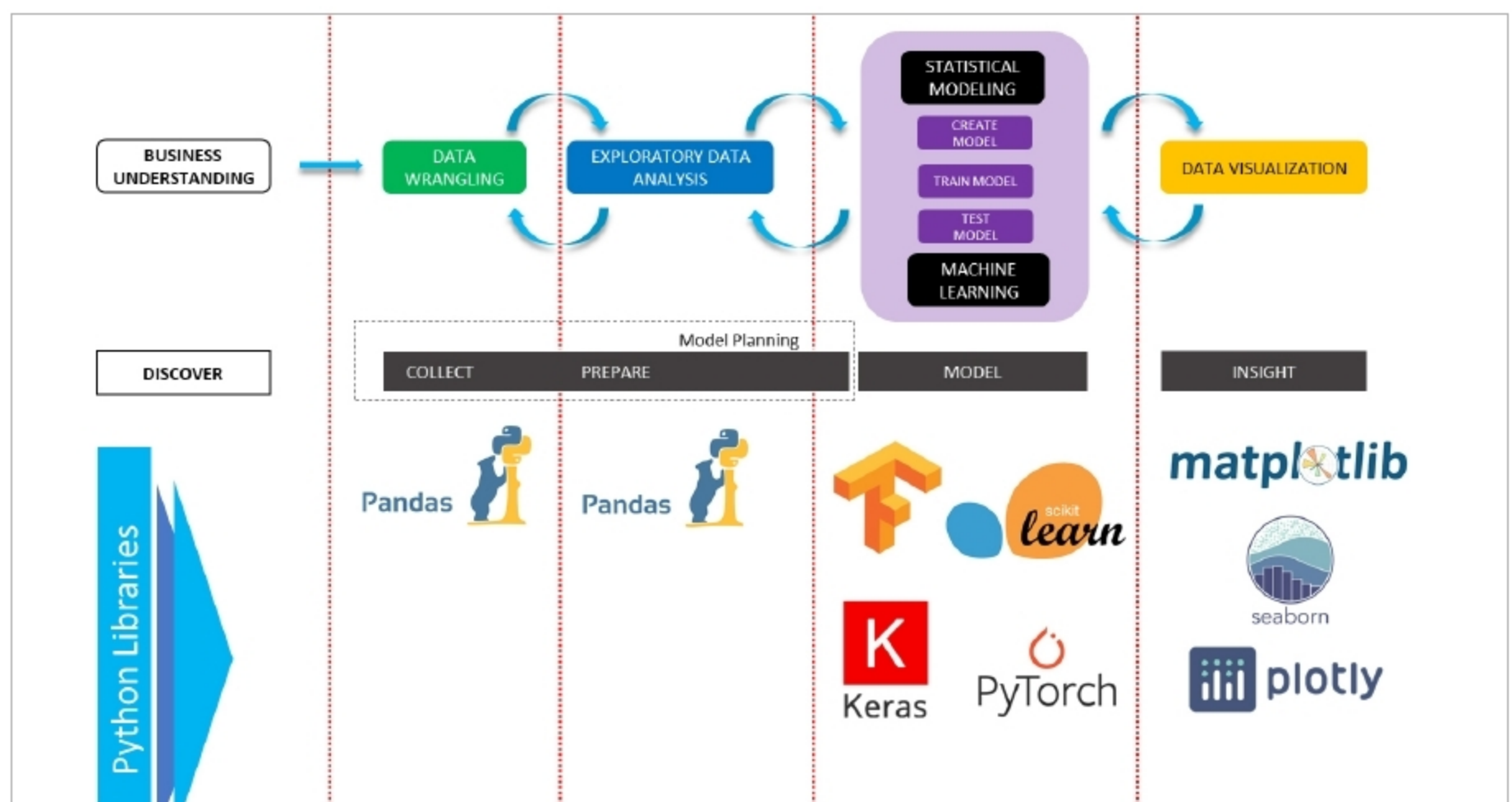


Figure 1: Python libraries for data science

